

The half-width multiplicity profile of a cylindric slice is *not* the positive part of a concave function and the sharp threshold is $n - m = 6$

Clio

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Abstract

Let $M(r) = \#\{\nu \in \Sigma_b^\circ : \Lambda(\nu) = r\}$ be the half-width multiplicity profile of a slice of a cylindric skew Kostka number, as in `2026-10-02-m3-concentricity.tex`, Remark `(rem:Mconcave)`. That remark records that M is the positive part of a concave function on all 135 422 nonzero slices with $2 \leq m \leq 5$, $m \leq n \leq m + 5$, $d \leq 11$, and names the general statement as “the first thing I would try to prove next”. **The general statement is false.** The smallest counterexample is $m = 2$, $n = 8$, $\mu = (0, 4)$, $\lambda = (3, 9)$, $b = 3$, where $M = (1, 1, 2)$ — not concave, and not even log-concave. The failure is one step outside the range of the discovery census: with $G := n - m$,

M is the positive part of a concave function for every $m \geq 2$, every cylindric $\mu \subseteq \lambda$ and every slice whenever $G \leq 5$; and for every $m \geq 2$ there is a counterexample with $G = 6$.

The discovery census covered exactly $G \leq 5$. Both halves are proved. The engine is one new identity: writing $a_i = \lambda_{i-1} + 1$,

$$\Lambda(\nu) = \frac{n - m}{2} - \frac{1}{2} \sum_{i=1}^m |\nu_i - a_i|, \quad \text{equivalently} \quad c - \Lambda(\nu) = \sum_{i=1}^m (\lambda_{i-1} + 1 - \nu_i)_+,$$

so the half-width is affine in the ℓ^1 -distance from ν to a point of $\mathbb{Z}^m$ fixed by λ alone, and its defect below the common centre $c = (d - b)/2$ is *the total failure of λ/ν to be a horizontal strip*. This identity re-proves parts (1) and (3) of the tent lemma in two lines, reduces M to the ℓ^1 -sphere counting function of a box-slice, and makes the threshold $G = 6$ a finite computation uniform in m . The failure does not threaten condition (A): on every counterexample found, $k(\cdot, b)$ is still PF₂.

1 Statement, and what is not being claimed

Notation is that of `2026-10-02-m3-concentricity.tex` (cited below as [c3]), in turn following [c1, Prop. 2.3]. Fix $m \geq 1$, a level $n \geq m$, cylindric shapes $\mu \subseteq \lambda$ in $\mathcal{C}^{n,m}$ stored as bead positions with $\mu_{i+m} = \mu_i + n$ and $\lambda_{i+m} = \lambda_i + n$, and put $d = |\lambda/\mu|$. With

$$L_i(\nu) = \max(\nu_i, \lambda_{i-1} + 1), \quad R_i(\nu) = \min(\nu_{i+1} - 1, \lambda_i), \quad w_i(\nu) = R_i(\nu) - L_i(\nu) + 1,$$

indices cyclic, the slice $\Sigma_b = \{\nu \in \mathcal{B}(\mu) : S_\nu = |\mu| + b\}$ carries $\Sigma_b^\circ = \{\nu \in \Sigma_b : f_\nu \neq 0\}$ and

$$\Lambda(\nu) = \frac{1}{2} \left(\sum_i w_i(\nu) - m \right), \quad M(r) = \#\{\nu \in \Sigma_b^\circ : \Lambda(\nu) = r\}.$$

By [c3, Thm. (thm:C)] every f_ν on the slice is symmetric about the common centre $c = (d - b)/2$, so $\Lambda(\nu)$ really is the half-width of the summand f_ν and M really is a summary of the slice.

Throughout, for a sequence $v \geq 0$ on $\mathbb{Z}$, “ v is the positive part of a concave function” means $v = c_+$ pointwise for some concave $c : \mathbb{R} \rightarrow \mathbb{R}$.

Lemma 1. *Let $v : \mathbb{Z} \rightarrow \mathbb{Z}_{\geq 0}$ have finite support. Then $v = c_+$ for some concave $c : \mathbb{R} \rightarrow \mathbb{R}$ if and only if the support of v is an interval $[r_0, r_1]$ and $v(r - 1) + v(r + 1) \leq 2v(r)$ for every $r_0 < r < r_1$.*

Proof. ($\Rightarrow$) If $v = c_+$ with c concave, then $\{c > 0\}$ is an interval, being a strict superlevel set of a concave function, so the support of v is a set of consecutive integers; and for $r_0 < r < r_1$ the three values $v(r - 1), v(r), v(r + 1)$ equal $c(r - 1), c(r), c(r + 1)$, so the inequality is concavity of c at r .

($\Leftarrow$) Let c_0 be the piecewise-linear interpolation of v on $[r_0, r_1]$; it is concave there precisely because the increments $v(r + 1) - v(r)$ are nonincreasing. Put

$$s_- = \max(v(r_0), v(r_0 + 1) - v(r_0)), \quad s_+ = \max(v(r_1), v(r_1 - 1) - v(r_1))$$

(with the convention $v(r_0 + 1) = v(r_1 - 1) = 0$ if $r_0 = r_1$), and extend c_0 to $c : \mathbb{R} \rightarrow \mathbb{R}$ by the line of slope s_- through $(r_0, v(r_0))$ on $(-\infty, r_0]$ and the line of slope $-s_+$ through $(r_1, v(r_1))$ on $[r_1, \infty)$. The slopes of c are nonincreasing left to right — $s_- \geq v(r_0 + 1) - v(r_0)$ and $-s_+ \leq v(r_1) - v(r_1 - 1)$ by construction — so c is concave. For an integer $r \leq r_0 - 1$, $c(r) = v(r_0) - s_-(r_0 - r) \leq v(r_0) - s_- \leq 0$ since $s_- \geq v(r_0)$, and symmetrically on the right. Hence $c_+ = v$. $\square$

So for the sequences at hand the property is exactly *interval support plus concavity on the support*. Interval support is [c3, Thm. (thm:tent)(3)]; the content is concavity.

Scope, stated before the proof and not after. [c3, Thm. (thm:blind)] is proved: the two multisets $\{(1, 1), (2, 2), (1, 5), (2, 6)\}$ and $\{(1, 1), (2, 2), (3, 3), (4, 4)\}$ have the *same* profile $M = (1, 1, 1, 1)$ — the flattest and most concave profile available — and sum to $(1, 3, 4, 6, 4, 3, 1) \notin \text{PF}_2$ and $(1, 3, 6, 10, 6, 3, 1) \in \text{PF}_2$ respectively. Consequently **no statement about M alone, true or false, can close condition (A) at $m \geq 3$** . Nothing below bears on (A). What is settled below is whether M -concavity is a genuine invariant of the family, which [c3, Rem. (rem:Mconcave)] asserted and which was the stated target. It is not.

2 The half-width is an ℓ^1 distance

Write $a_i = \lambda_{i-1} + 1$ for $1 \leq i \leq m$ (cyclically, so $a_1 = \lambda_m - n + 1$), and $(t)_+ = \max(t, 0)$, $\neg t = \max(-t, 0)$.

Introduce, once and for all,

$$y_i = \nu_i - a_i, \quad g_i = \lambda_i - \lambda_{i-1} - 1 \quad (\geq 0), \quad G := \sum_{i=1}^m g_i = n - m, \quad (1)$$

the last equality by the cyclic telescoping $\sum_i \lambda_{i-1} = |\lambda| - n$ noted below.

Lemma 2 (the width vector in closed form). *For every $\nu \in \mathbb{Z}^m$ and every i ,*

$$w_i(\nu) = g_i - (y_{i+1})_- - (y_i)_+ + 1. \quad (2)$$

Proof. $\max(\nu_i, \lambda_{i-1} + 1) = a_i + \max(y_i, 0) = a_i + (y_i)_+$, and $\nu_{i+1} - 1 = a_{i+1} + y_{i+1} - 1 = \lambda_i + y_{i+1}$, so $\min(\nu_{i+1} - 1, \lambda_i) = \lambda_i + \min(y_{i+1}, 0) = \lambda_i - (y_{i+1})_-$. Subtract and add 1: $w_i = (\lambda_i - \lambda_{i-1} - 1) - (y_{i+1})_- - (y_i)_+ + 1$. $\square$

Equation (2) is worth pausing on. The width vector sees $(y_i)_+$ and $(y_i)_-$ *separately*; the half-width, being $\frac{1}{2}(\sum_i w_i - m)$, sees only their sum $|y_i|$. That is exactly the information that (thm:blind) says a proof of (A) must have and a statement about M cannot. It also gives Theorem 3 immediately.

Theorem 3. *For every $m \geq 1$, every cylindric $\mu \subseteq \lambda$ in $\mathcal{C}^{n,m}$ and every $\nu \in \mathbb{Z}^m$,*

$$\Lambda(\nu) = \frac{n-m}{2} - \frac{1}{2} \sum_{i=1}^m |\nu_i - a_i|. \quad (3)$$

On the slice $S_\nu = |\mu| + b$, with $c = (d-b)/2$,

$$c - \Lambda(\nu) = \sum_{i=1}^m (a_i - \nu_i)_+ = \sum_{i=1}^m (\lambda_{i-1} + 1 - \nu_i)_+ =: k(\nu) \geq 0, \quad (4)$$

and $k(\nu) = 0$ if and only if λ/ν is a horizontal strip.

Proof. Summing (2) over i and using $(t)_+ + (t)_- = |t|$,

$$\sum_{i=1}^m w_i(\nu) = G - \sum_{i=1}^m |y_i| + m, \quad \text{so} \quad \Lambda(\nu) = \frac{\sum_i w_i - m}{2} = \frac{G}{2} - \frac{1}{2} \sum_{i=1}^m |y_i|,$$

which is (3). (The cyclic convention $\lambda_0 = \lambda_m - n$ gives $\sum_{i=1}^m \lambda_{i-1} = \sum_{j=0}^{m-1} \lambda_j = |\lambda| - n$, whence $\sum_i g_i = |\lambda| - (|\lambda| - n) - m = n - m = G$, as used in (1); this cyclic closure is the only place the cylindricity of λ enters.) The same substitution in [c3]'s formula (eq:Lam) gives (3) a second time, since $(t)_+ = \frac{1}{2}(t + |t|)$ turns $\sum_i (\nu_i - a_i)_+$ into $\frac{1}{2}(S_\nu - A) + \frac{1}{2} \sum_i |y_i|$ with $A = \sum_i a_i = |\lambda| - n + m$, and the S_ν terms cancel.

For (4) put $\sigma = \sum_i y_i$. On the slice

$$\sigma = S_\nu - A = |\mu| + b - |\lambda| + n - m = n - m - d + b, \quad (5)$$

and $\sum_i |y_i| = \sum_i y_i + 2 \sum_i (y_i)_- = \sigma + 2k(\nu)$, so

$$\Lambda(\nu) = \frac{G}{2} - \frac{\sigma}{2} - k(\nu) = \frac{(n-m) - (n-m-d+b)}{2} - k(\nu) = \frac{d-b}{2} - k(\nu) = c - k(\nu).$$

Finally $k(\nu) = 0$ iff $\nu_i \geq \lambda_{i-1} + 1$ for every i ; together with $\nu_i \leq \lambda_i$ (which holds on Σ_b° , since $R_i \geq L_i$ forces it) this is exactly the bead-coordinate criterion $\lambda_{i-1} < \nu_i \leq \lambda_i$ for λ/ν to be a horizontal strip. $\square$

Two parts of the tent lemma now cost two lines each.

Corollary 4. Λ is concave on $\mathbb{R}^m$, being $\frac{n-m}{2}$ minus half a norm. Each unit step in one coordinate changes Λ by exactly $\pm \frac{1}{2}$, so along an exchange $\nu \mapsto \nu + e_i - e_j$ one has $|\Delta \Lambda| \leq 1$. On a slice $\Lambda \leq c$, with equality exactly when λ/ν is a horizontal strip, and $\Lambda \in c - \mathbb{Z}_{\geq 0}$.

Remark 5 (where the two inputs are spent). The brief asked where each quoted hypothesis is consumed. **Theorem (thm:tent) is spent twice**: its formula (eq:Lam) is the input to Theorem 3, and its part (2) (the effective index set is a box) is the input to §3. Its parts (1) and (3) are *not* spent — Corollary 4 recovers them. **Theorem (thm:C) is not spent at all**. It is what makes M a meaningful summary of a slice (all summands concentric) and it supplies the *value* $c = (d - b)/2$ that turns k into a defect below a common centre; but the assertion “ M is the positive part of a concave function” is a statement about Λ alone, and nothing below uses concentricity. By the rule that a hypothesis which is never consumed should be dropped, the right ambient statement of today’s question never mentioned (C).

3 Reduction to a box-slice, and the two sharp bounds

With y_i, g_i, G as in (1), put $\delta_i = \mu_i - \lambda_{i-1} - 1$. Since $\mu_i \leq \lambda_i$ we have $\delta_i \leq g_i$, and $\mu_i < \mu_{i+1}$ reads $\delta_{i+1} \geq \delta_i - g_i$.

Proposition 6. Σ_b° is carried by y bijectively onto

$$\left\{ y \in \prod_{i=1}^m [P_i, Q_i] : \sum_i y_i = \sigma \right\}, \quad P_i = \max(\delta_i, -g_{i-1}), \quad Q_i = g_i - (\delta_{i+1})_-,$$

with $\sigma = n - m - d + b = \sum_i \delta_i + b$ and $d = G - \sum_i \delta_i$; and $\Lambda = c - k$ with $k(y) = \sum_i (y_i)_-$. Writing $u_i = Q_i - P_i$, $X = \sum_i (\delta_i)_-$ and $Y = \sum_i (\delta_i + g_{i-1})_+$,

$$\sum_{i=1}^m u_i = 2G - X - Y, \quad (\text{B1}) \quad \sum_{i=1}^m u_i + \sum_{i=1}^m (\delta_i)_+ \leq G, \quad (\text{B2}) \quad \sum_{i=1}^m Q_i + \sum_{i=1}^m (P_i)_- \leq G. \quad (6)$$

Consequently $\sum_i (Q_i)_+ \leq G$, $\sum_i (P_i)_- \leq G$, $P_i \in [-G, G]$, $\#\{i : u_i > 0\} \leq G$, and $d \leq 3G$ whenever the slice is nonempty.

Proof. The box is $[\text{c3}, (\text{eq:effbox})]$ translated by a : $\mathcal{B}_i^\circ = [\max(\mu_i, \lambda_{i-2} + 2), \min(\mu_{i+1} - 1, \lambda_i)]$, and $\max(\mu_i, \lambda_{i-2} + 2) - a_i = \max(\delta_i, -g_{i-1})$ because $\lambda_{i-2} + 2 - \lambda_{i-1} - 1 = -g_{i-1}$, while $\min(\mu_{i+1} - 1, \lambda_i) - a_i = \min(\delta_{i+1} + g_i, g_i) = g_i - (\delta_{i+1})_-$ because $\mu_{i+1} - 1 - \lambda_{i-1} - 1 = \delta_{i+1} + g_i$. The value of σ is (5), and $d = \sum_i (\lambda_i - \mu_i) = \sum_i (g_i - \delta_i) = G - \sum_i \delta_i$.

For (6), use $\max(s, t) = t + (s - t)_+$ to write $P_i = -g_{i-1} + (\delta_i + g_{i-1})_+$, so

$$u_i = (g_i - (\delta_{i+1})_-) + g_{i-1} - (\delta_i + g_{i-1})_+, \quad \text{hence} \quad \sum_i u_i = 2G - X - Y.$$

Now $(\delta_i + g_{i-1})_+ \geq g_{i-1} - (\delta_i)_- + (\delta_i)_+$ for every i : if $\delta_i \geq 0$ both sides equal $\delta_i + g_{i-1}$; if $\delta_i < 0$ the right side is $g_{i-1} + \delta_i \leq (\delta_i + g_{i-1})_+$. Summing, $Y \geq G - X + \sum_i (\delta_i)_+$, and substituting into $\sum u_i = 2G - X - Y$ gives (B1). For (B2), $\sum_i Q_i = G - X$, and $(P_i)_- = g_{i-1} - (\delta_i + g_{i-1})_+$ when $\delta_i \leq 0$ while $(P_i)_- = 0$ when $\delta_i > 0$, so

$$\sum_i (P_i)_- = \sum_{\delta_i \leq 0} g_{i-1} + \sum_{\delta_i > 0} (\delta_i + g_{i-1}) - Y = G + \sum_i (\delta_i)_+ - Y,$$

whence $\sum_i Q_i + \sum_i (P_i)_- = 2G - X - Y + \sum_i (\delta_i)_+ = \sum_i u_i + \sum_i (\delta_i)_+ \leq G$ by (B1).

The consequences: $P_i \leq Q_i$ gives $(P_i)_- \geq (Q_i)_-$, so (B2) yields $\sum_i (Q_i)_+ = \sum_i Q_i + \sum_i (Q_i)_- \leq G$; $\sum_i (P_i)_- \leq G$ is (B2) with $\sum_i Q_i \geq -\sum_i (P_i)_-$ absorbed, more directly $(P_i)_- \leq g_{i-1}$ summed; $P_i \geq -G$ and $Q_i \leq G$ follow; $u_i \geq 1$ on at most G indices by (B1); and $\sum_i u_i \geq 0$ with (6) gives $X \leq 2G$, i.e. $\sum_i \delta_i \geq -2G$, i.e. $d = G - \sum_i \delta_i \leq 3G$. $\square$

So M is, after the affine reindexing $r = c - k$ of Theorem 3, the ℓ^1 -sphere counting function

$$N(k) = \#\left\{y \in \prod_i [P_i, Q_i] : \sum_i y_i = \sigma, \sum_i (y_i)_- = k\right\}, \quad M(c - k) = N(k). \quad (7)$$

Concavity is invariant under $r \mapsto c - r$, so $M = c_+$ iff N is the positive part of a concave function.

Remark 7 (the abstract box class is not enough, and that is the point). Proposition 6 does *not* reduce the question to arbitrary boxes. Over all boxes $\prod_i [P_i, Q_i] \subset \mathbb{Z}^m$ with $P_i, Q_i \in [-3, 3]$ and all σ , N fails to be the positive part of a concave function on 58 of 3 920 cases at $m = 2$, 5 576 of 153 664 at $m = 3$, and 440 168 of 5 531 904 at $m = 4$; the smallest is $P = (-3, -3)$, $Q = (0, 2)$, $\sigma = -1$, where $N = (2, 1, 1)$. The bounds (B1),(B2) are therefore load-bearing, and §5 shows they are also *sharp*: they are exactly what makes $G \leq 5$ clean and $G = 6$ dirty.

4 $m = 2$: the exact structure, and the mechanism of failure

Proposition 8. *Let $m = 2$ and let $\Sigma_b^\circ \neq \emptyset$. Put $A = \max(P_1, \sigma - Q_2)$, $Z = \min(Q_1, \sigma - P_2)$, so that y_1 ranges over the interval $[A, Z]$, and put $I_0 = [\min(0, \sigma), \max(0, \sigma)]$. Then*

$$k(y) = (\sigma)_- + \text{dist}(y_1, I_0). \quad (8)$$

Set $F = \#([A, Z] \cap I_0)$, $L = \#\{t \in [A, Z] : t < \min I_0\}$ and $R = \#\{t \in [A, Z] : t > \max I_0\}$. If $F = 0$ then $N = (1, 1, \dots, 1)$. If $F \geq 1$ then, with $J = \min(L, R)$ and $D = |L - R|$,

$$N = (F, \underbrace{2, \dots, 2}_J, \underbrace{1, \dots, 1}_D). \quad (9)$$

Consequently M is not the positive part of a concave function exactly when $F \geq 1$ and one of

- (i) $J = 0, D \geq 2, F \geq 2$; (ii) $J \geq 1, D \geq 2$;
- (iii) $J \geq 2, D \leq 1, F \geq 3$; (iv) $J = 1, D = 1, F \geq 4$.

Proof. With $y_2 = \sigma - y_1$ we have $k = (y_1)_- + (\sigma - y_1)_- = (-y_1)_+ + (y_1 - \sigma)_+$. If $\sigma \geq 0$ this is $-y_1$ for $y_1 < 0$, 0 on $[0, \sigma]$, and $y_1 - \sigma$ for $y_1 > \sigma$, i.e. $\text{dist}(y_1, [0, \sigma])$, and $(\sigma)_- = 0$. If $\sigma < 0$ it is $-y_1$ for $y_1 < \sigma$, $-\sigma$ on $[\sigma, 0]$, and $y_1 - \sigma$ for $y_1 > 0$, i.e. $-\sigma + \text{dist}(y_1, [\sigma, 0])$. This is (8).

$[A, Z]$ is an interval. If it misses I_0 it lies entirely on one side of I_0 , so $\text{dist}(\cdot, I_0)$ is injective on it and N is a string of 1s. If it meets I_0 then $A \leq \max I_0$ and $Z \geq \min I_0$, so $\{t \in [A, Z] : t < \min I_0\} = [A, \min I_0 - 1]$ realises the distances $1, \dots, L$ once each, and $\{t \in [A, Z] : t > \max I_0\} = [\max I_0 + 1, Z]$ realises $1, \dots, R$ once each; distance 0 is realised F times. This is (9).

For the criterion, apply Lemma 1 to (9). Its increment sequence is $(2 - F, 0^{J-1}, -1, 0^{D-1})$ if $J \geq 1, D \geq 1$; $(2 - F, 0^{J-1})$ if $J \geq 1, D = 0$; $(1 - F, 0^{D-1})$ if $J = 0, D \geq 1$; and empty if $J = D = 0$. Nonincreasingness fails precisely in the four listed cases: an interior -1 followed by a 0 (case ii, and case i with $J = 0$ reading $1 - F < 0$), or a leading increment below the next one (cases iii and iv). $\square$

The mechanism is now visible. M at $m = 2$ is governed by three integers: the length F of the flat bottom of the tent k , and the lengths L, R of its two *arms* after truncation by the box. Concavity of M is a *balance* condition on the two arms — and nothing in [c3] couples them. The $m = 2$ proof of (A) in [c2] never needed this balance, because it spent concavity one level down, at the layer profile β ; and the layer-level statement is false at $m \geq 3$ ([c3, Prop. (prop:dead)]). Remark (**rem:Mconcave**) proposed that the concavity had *relocated* one level up. It had not. It is absent at both levels; at the upper level it merely takes more room to see.

Example 9 (the smallest counterexample). $m = 2$, $n = 8$, $\mu = (0, 4)$, $\lambda = (3, 9)$, $d = 8$, $b = 3$, so $c = \frac{5}{2}$. Here $\lambda_0 = 1$, $g = (1, 5)$, $\delta = (-2, 0)$, hence $P = (-2, 0)$, $Q = (1, 3)$, $\sigma = 1$. The effective slice and its summands are

ν	$(L_i, R_i)_i$	$w(\nu)$	f_ν	$\Lambda(\nu)$	k
$(0, 7)$	$(2, 3), (7, 7)$	$(2, 1)$	$(1, 1)$ on $[2, 3]$	$\frac{1}{2}$	2
$(1, 6)$	$(2, 3), (6, 8)$	$(2, 3)$	$(1, 2, 2, 1)$ on $[1, 4]$	$\frac{3}{2}$	1
$(2, 5)$	$(2, 3), (5, 9)$	$(2, 5)$	$(1, 2, 2, 2, 2, 1)$ on $[0, 5]$	$\frac{5}{2}$	0
$(3, 4)$	$(3, 3), (4, 9)$	$(1, 6)$	$(1, 1, 1, 1, 1, 1)$ on $[0, 5]$	$\frac{5}{2}$	0

All four are concentric about $c = \frac{5}{2}$, as Theorem (thm:C) requires. Hence

$$M = (1, 1, 2) \quad \text{at } r = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \quad N = (2, 1, 1) \quad \text{at } k = 0, 1, 2,$$

with increments $0, +1$ in r : **not concave**, and since $1^2 < 1 \cdot 2$, **not even log-concave**. In the language of Proposition 8: $A = -2$, $Z = 1$, $I_0 = [0, 1]$, so $F = 2$, $L = 2$, $R = 0$, i.e. $J = 0$, $D = 2$, $F = 2$ — case (i), the minimal failure. The two arms are maximally unbalanced: one has length 2, the other is empty. The slice sum is nevertheless $k(\cdot, 3) = (2, 4, 6, 6, 4, 2)$, which is PF_2 .

5 The sharp threshold $G = n - m = 6$

Theorem 10. *Let $G = n - m$.*

1. *If $G \leq 5$ then for every $m \geq 1$, every cylindric $\mu \subseteq \lambda$ in $\mathcal{C}^{n,m}$ and every b , the profile M is the positive part of a concave function.*
2. *For every $m \geq 2$ this is sharp: with $G = 6$, i.e. $n = m + 6$, and*

$$\begin{aligned} m = 2: \quad & \mu = (0, 4), & \lambda = (3, 9); \\ m \geq 3: \quad & \mu = (0, 1, \dots, m-2, m+2), & \lambda = (0, 1, \dots, m-3, m+3, m+5), \end{aligned}$$

one has $d = 8$ and, at $b = 3$, $M = (1, 1, 2)$, which is not the positive part of a concave function and is not log-concave.

Proof. (2). For $m \geq 3$ one computes directly from the displayed shapes: $\lambda_0 = -1$,

$$g = (0, \dots, 0, 5, 1), \quad \delta = (0, \dots, 0, -2), \quad \sum_i g_i = 6 = G, \quad d = G - \sum_i \delta_i = 8.$$

By Proposition 6, $P_i = \max(\delta_i, -g_{i-1}) = 0$ for $i \leq m-1$ and $P_m = \max(-2, -5) = -2$; $Q_i = g_i - (\delta_{i+1})_- = 0$ for $i \leq m-2$, $Q_{m-1} = 5 - 2 = 3$, $Q_m = 1 - 0 = 1$; and $\sigma = \sum_i \delta_i + b = -2 + 3 = 1$. So coordinates $1, \dots, m-2$ are frozen at 0 and the effective slice is

$$\{(0, \dots, 0, y_{m-1}, y_m) : y_{m-1} \in [0, 3], y_m \in [-2, 1], y_{m-1} + y_m = 1\} = \{(0, 1), (1, 0), (2, -1), (3, -2)\}$$

in the last two coordinates, with $k = 0, 0, 1, 2$ respectively. Hence $N = (2, 1, 1)$ and, since $c = (d - b)/2 = \frac{5}{2}$, $M = (1, 1, 2)$ at $r = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}$, whose increments $0, +1$ are not nonincreasing (Lemma 1) and which violates $1^2 \geq 1 \cdot 2$. The case $m = 2$ is Example 9, the same data read cyclically.

(1). Call i *frozen* if $u_i = Q_i - P_i = 0$, say $P_i = Q_i = p_i$. A frozen coordinate contributes the constant p_i to $\sum_i y_i$ and the constant $(p_i)_-$ to k , so by (7)

$$N(k) = N^T\left(k - \sum_{i \notin T} (p_i)_-\right), \quad T = \{i : u_i > 0\},$$

where N^T is the profile of the box $\prod_{i \in T} [P_i, Q_i]$ at $\sigma^T = \sigma - \sum_{i \notin T} p_i$. A translation in k does not affect concavity, so $M = c_+$ if and only if N^T is the positive part of a concave function. If $T = \emptyset$ then N is a single positive value and there is nothing to prove.

By Proposition 6, the reduced data satisfies: $\#T \leq G$; $u_i \geq 1$ and $\sum_{i \in T} u_i \leq G$; $P_i \geq -G$ and $Q_i \leq G$ for $i \in T$; and, because every frozen coordinate contributes $Q_i + (P_i)_- = p_i + (p_i)_- = (p_i)_+ \geq 0$ to the left side of (B2),

$$\sum_{i \in T} Q_i + \sum_{i \in T} (P_i)_- \leq G.$$

Finally $\sigma^T \in [\sum_{i \in T} P_i, \sum_{i \in T} Q_i]$, since the reduced slice is nonempty. These conditions are *necessary*, so the set of reduced data they cut out is a superset of what cylindric shapes can realise — and, for fixed G , it is finite and independent of m . Exhausting it for $G \leq 5$ and evaluating (7) directly gives no failure:

G	0	1	2	3	4	5
reduced (box, σ^T) cases	0	4	44	499	6 679	105 443
failures	0	0	0	0	0	0

(The computation is `bounds.py`; see §6, including the positive control that the same enumerator *does* return failures at $G = 6$.) Hence no cylindric datum with $G \leq 5$ can fail. $\square$

Corollary 11. *Remark (`rem:Mconcave`) of [c3] is correct as an empirical statement and false as a conjecture. Its census range was $m \leq n \leq m + 5$, i.e. exactly $G \leq 5$; by Theorem 10(1) a clean sweep there was guaranteed in advance, for every m and every d , and by Theorem 10(2) the conjecture fails at the next value of G , for every m .*

Remark 12 (what the census could not have seen). The deeper reason the 135 422 cases carried no information about concavity is visible in the profile lengths. Over $2 \leq m \leq 5$, $n \leq m + 5$, $d \leq 9$ every profile M has length ≤ 3 , and 90% have length ≤ 2 ; a sequence of length ≤ 2 is concave automatically, so for those slices the test has no content at all. Concavity first has two independent increments to compare at length 3, and the first genuine obstruction needs the asymmetric truncation of Proposition 8, which needs room. The census was not merely unlucky in its range: in its range the predicate was close to vacuous.

6 Census and controls

Code in `scratch/1003-prove/`. No instrument's output was believed before it reproduced a value already held, and no absence was believed before the same instrument was shown to detect a planted presence.

Calibration.

- The slice engine (`gen.py` of `proofs/code-1002-m3/`, unmodified) reproduces both held $m = 2$ values exactly: $k(\cdot, 2) = (1, 3, 6, 7, 6, 3, 1)$ at $n = 8$, $\mu = (0, 3)$, $\lambda = (4, 7)$, and $G = (1, 4, 7, 10, 11, 10, 7, 4, 1)$ at $n = 10$, $\mu = (0, 5)$, $\lambda = (5, 12)$, $b = 4$.
- The concave-positive-part test refuses $(3, 6, 6, 3, 1)$ and refuses $(2, 1, 1)$.
- **Theorem 3** is exact on all 202 188 ν tested ($2 \leq m \leq 6$, $n \leq m + 4$, $d \leq 7$), and agrees with the *measured* half-width $(\text{len } f_\nu - 1)/2$ on all 89 377 nonzero summands; zero disagreements. Equation (4) holds on all 550 408 summands of 269 289 slices ($2 \leq m \leq 5$, $n \leq m + 6$, $d \leq 8$), and $k(\nu) = 0 \Leftrightarrow \lambda/\nu$ a horizontal strip agrees with an independent `hstrip` test on the same 550 408; zero disagreements.
- **Proposition 6** (P_i, Q_i, σ, d in terms of g, δ , and the identities and bounds (6)) is exact on all 115 189 nonempty boxes tested; the y -box description of Σ_b° agrees setwise with direct enumeration on all 29 341 nonzero slices tested; $d \leq 3G$ holds on all 12 295 nonempty cases tested.
- **Proposition 8** (both (9) and the four-case criterion) is exact over all 74 529 pairs (box, σ) with $m = 2$ and $P_i, Q_i \in [-6, 6]$, checked against brute-force enumeration; zero disagreements.

The refutation, by three independent routes.

- **Route 1 (yesterday’s own instrument).** `mprofile.py` of `proofs/code-1002-m3/`, with the single change $n \in [m, m + 5] \mapsto n \in [m, m + 9]$ and nothing else: at $m = 2$, $d \leq 11$, it reports 10 185 slices with 100 failures of “ $M = c_+$ ” and the same 100 failures of log-concavity, first witness $(m, n, \mu, \lambda, b) = (2, 8, (0, 4), (3, 9), 3)$, $M = (1, 1, 2)$. Run in its original range ($m \in \{2, 3, 4\}$, $n \leq m + 5$, $d \leq 11$) it reports 42 718 slices and *zero* failures, reproducing [c3].
- **Route 2 (raw supports).** An instrument that never forms the y -box and never uses (3) — it reads half-widths off $\text{len } f_\nu$ built from `gen.LR/gen.f_nu` — finds the first failing level at $n = 8$ with four witnesses, and reproduces the table of Example 9 entry by entry.
- **Route 3 (box engine).** The generating-function profile of Proposition 6 agrees with Routes 1 and 2 and supports the wide census below.

Extended census. Via Route 3, over (μ, λ, b) with $\mu_1 = 0$:

m	range	slices	failures	max profile length
2	$n \leq 20, d \leq 24$	259 652	21 380	9
3	$n \leq 13, d \leq 16$	323 852	6 531	6
4	$n \leq 12, d \leq 14$	629 789	2 630	5
5	$n \leq 12, d \leq 13$	1 069 479	950	4

Sweeps at $m = 6, 7$ were launched over $n \leq 13$, $d \leq 13$ and $n \leq 14$, $d \leq 13$ and had not returned at write-up. **This is a deliberately reported bound on coverage, not a silent cap** — but it costs nothing, because Theorem 10 covers both m exactly: part (1) is uniform in m , so no failure is possible at $m = 6, 7$ with $G \leq 5$, and part (2) was verified in closed form at $m = 2, 3, \dots, 12$, giving $M = (1, 1, 2)$ and $k(\cdot, 3) \in \text{PF}_2$ in every case. The missing rows would

have been confirmatory only. Indeed the same observation indicts the shape of the original census: a range written as $m \leq n \leq m + 5$ pins $G \leq 5$ *uniformly in m* , so enlarging m — which looks like broadening coverage — buys nothing at all. The governing parameter is $G = n - m$, and no amount of m substitutes for one more unit of it.

Threshold, exhaustively. Because $d \leq 3G$ on every nonempty slice (Proposition 6), sweeping $d \leq 3G$ at fixed (m, n) is *complete*. Doing so:

$G = n - m$	0	1	2	3	4	5	6	7
$m = 2$	1/0	12/0	60/0	192/0	475/0	996/0	1 862/4	3 200/24
$m = 3$	1/0	18/0	138/0	652/0	2 265/0	6 384/0	15 484/12	33 552/126
$m = 4$	1/0	24/0	248/0	1 568/0	7 140/0	25 752/0	78 132/24	207 576/384
$m = 5$	1/0	30/0	390/0	3 100/0	17 595/0	78 084/0	287 405/40	—

Entries are *slices/failures*; each row is complete, since $d \leq 3G$. In every case the first witness is the reduced box $P = (0, \dots, 0, -2)$, $Q = (0, \dots, 0, 3, 1)$, $\sigma = 1$, $N = (2, 1, 1)$ of Theorem 10(2).

Controls.

1. **Positive control on the absence claim.** The finite enumerator of Theorem 10(1) is the instrument certifying an absence, so it was first shown to detect a presence: at $G = 6$ it returns 10 failures out of 1 931 110 reduced cases. An enumerator that returned 0 at $G = 6$ would have invalidated the $G \leq 5$ rows regardless of their arithmetic.
2. **Refusal control: are (B1) and (B2) load-bearing and tight?** Decouple the two budgets, allowing $\sum_i u_i \leq U$ and $\sum_i Q_i + \sum_i (P_i)_- \leq B$ independently. Prediction, read off Example 9 (where $\sum u_i = 6$ and $\sum Q_i + \sum (P_i)_- = 1 + 3 + 2 + 0 = 6$): the least admissible budgets are $U = 6$ and $B = 6$, neither alone. Outcome:

U	B	cases	verdict
5	5	105 443	clean
6	5	202 858	clean
5	6	325 478	clean
6	6	1 931 110	fails: $P = (-6, -1)$, $Q = (-3, 2)$, $\sigma = -4$, $N = (2, 1, 1)$
8	5	609 932	clean
5	8	1 690 934	clean

Neither bound is surplus, and neither can be relaxed alone even generously: holding $U = 5$ one may push B to 8 and stay clean, and holding $B = 5$ one may push U to 8 and stay clean. Both budgets must reach 6 simultaneously. The threshold $G = 6$ is therefore not an artefact of an enumeration range: it is exactly the point at which (B1) and (B2) jointly stop excluding $N = (2, 1, 1)$.

3. **Which hypotheses are spent.** `(thm:tent)(eq:Lam)` and `(thm:tent)(2)` are each spent once; `(thm:tent)(1)` and (3) are recovered as Corollary 4 and are therefore surplus; `(thm:C)` is spent nowhere. Four perturbations of `(eq:Lam)` and `(eq:effbox)` are refused on 229/466, 466/466, 400/1047, 434/1047 instances (inherited from [c3]).
4. **Abstract-class control** (Remark 7). The statement is *false* in the abstract class “box $\cap$ hyperplane”, so the real constraints are doing work and the result is not a theorem about lattice points in general position.

7 What this leaves

1. **Condition (A) is untouched.** On every counterexample computed — including the whole $m = 2, \dots, 12$ family of Theorem 10(2) and all 21 380 failures of the $m = 2$ census — $k(\cdot, b)$ remains PF_2 . The invariant that died was never on the critical path: `(thm:blind)` had already proved that no condition on M alone can suffice.
2. **The surviving positive content** is Theorem 3: the half-width is affine in the ℓ^1 -distance from ν to $a = (\lambda_{i-1} + 1)_i$, and its defect below the common centre is the horizontal-strip defect of λ/ν . That is a closed-form, shape-aware description of Λ — and `(thm:blind)` says that shape-awareness is exactly what any proof of (A) must have. And Lemma 2 says more than Theorem 3 does: the open problem of [c3] — “which m -tuples $w(\nu)$ co-occur in one slice” — now has coordinates,

$$w_i(\nu) = g_i - (y_{i+1})_- - (y_i)_+ + 1 \quad \text{on the box-slice of Proposition 6,}$$

exact on all 175 746 nonzero summands tested ($2 \leq m \leq 5$, $n \leq m + 5$, $d \leq 8$; zero disagreements, and $\sum_i w_i = 2\Lambda + m$ on the same set). The whole question is therefore: *for which (P, Q, σ) arising as in Proposition 6 does the multiset $\{w(y) : y \in \text{box-slice}\}$, with w as displayed, sum to a PF_2 sequence?* The half-widths discard the split of y_i into $(y_i)_+$ and $(y_i)_-$ and keep only $|y_i|$ — which is, precisely and quantitatively, the information `(thm:blind)` proves no proof of (A) can do without.

3. **A methodological residue.** The failure was not subtle and not deep: it sits one unit of n outside a census of 135 422 cases, and in that census the predicate being tested was nearly vacuous (90% of profiles had length ≤ 2). The lesson is not “sweep wider”. It is that a census should report the *length of the objects it tests*, because a predicate with no content returns green at any scale.

References. All in `~/projects/proofs/`.

[c1] 2026-09-30-c1-cylindric-kostka-logconcavity.tex, Prop. `prop:reform` (the slice-sum reformulation).

[c2] 2026-10-01-c2-inter-region-inequality.tex (the $m = 2$ proof of (A)).

[c3] 2026-10-02-m3-concentricity.tex.